Hypergraph Community Detection With Fuzzy Memberships

Jing Xiao, Zhi-Wei Ma, Jing Cao, and Xiao-Ke Xu, *Member, IEEE*

***Abstract*—Hypergraph community detection reveals both mesoscale structures and functional characteristics of real-life hypergraphs. Although many methods have been developed from diverse perspectives, to our knowledge, none can provide fine-grained hypergraph fuzzy community information, and the quality of identified crisp partitions is often limited. This study defines a set of novel concepts of multi-scale hypergraph fuzzy memberships that systematically quantify the partial belongingness (i.e., membership grades) among nodes, hyperedges, and hypergraph communities within the interval [0,1], thereby revealing the multi-scale partial topological affiliations. Furthermore, the multiscale hypergraph fuzzy memberships are employed to develop a general framework, named fuzzy membership-assisted hypergraph modularity optimization (FMHMO), aiming to approximate the modularity-optimal crisp hypergraph partition. The FMHMO framework comprises two key strategies: Hyperedge similarity-based hypergraph reduction (HS-HR) and fuzzy membership-based hypergraph partition recovery (FM-HPR). In particular, HS-HR reduces a hypergraph to a simple weighted graph, mapping hyperedges as nodes and their interactions as edges using novel hyperedge similarity as weights. Thereby, preserving intrahyperedge and interhyperedge topologies of the original hypergraph. FM-HPR recovers a hypergraph partition from the approximately modularity-optimal weighted graph partition by quantifying the fuzzy memberships of bridge nodes connecting multiple incident hyperedges, thereby enabling precise assignment of their crisp hypergraph community affiliations. Experimental results in both synthetic and real-world datasets indicate the superiority of FMHMO over state-of-the-art hypergraph community detection algorithms in accuracy and modularity quality.**

***Index Terms*—Hyperedge, hypergraph community detection, hypergraph fuzzy membership, modularity optimization.**

I. Introduction

HYPERGRAPHS are generalized graphs characterized by hyperedges, which are prevalent in representing complex real-world network systems across various domains, including social [1], biological [2] and engineering [3]. Community structure is one of the most important topological characteristics of real-life hypergraphs [4]. However, despite the development of numerous community detection methods based on diverse knowledge domains [5], [6], [7], most primarily focus on identifying communities within ordinary simple graphs [8]. Consequently, the important mesoscale structural characteristics of hypergraphs remain significantly underexplored [9].

In essence, hypergraph community detection aims to find dense connected components of nodes within a hypergraph [10], [11], promoting deep understanding of mesoscale structures and functional characteristics of real-world hypergraphs [12]. Specifically, node density and hyperedge-based interaction strength within a community are expected to be dense, whereas those between communities should be sparse [13]. However, unlike edges in ordinary graphs with pairwise (or dyadic) simple relationships between two nodes, hyperedges in hypergraphs represent the polyadic complex group interactions among more than two nodes [14], [15]. In addition, compared with ordinary higher-order connectivity patterns (e.g., network motifs), hyperedges are featured by arbitrary number of internal nodes, variable aggregations of internal nodes, and diverse interactions both between hyperedges and between hyperedges and communities. The complex and diversified higher-order topological structures of hyperedges make hypergraph community detection a challenging task [16].

In recent years, hypergraph community detection has garnered increasing attention, leading to the development of several representative methods from diverse perspectives. These include methods based on derivative graphs [13], probabilistic models [4], [16], modularity-based optimization [9], [11], etc. Particularly, modularity-based optimization methods exhibit competitive performance, due to the advantages of approximating the global optimum, stability, and few parameters [17], [18]. Essentially, they model hypergraph community detection as a global optimization problem with a specific modularity function, such as hypergraph modularities Q^{hyp} [19] and q_H [9]. In general, existing hypergraph community detection methods have demonstrated effectiveness in identifying crisp hypergraph partitions of nodes, revealing higher-order organizational structures within real-world hypergraphs. Despite their effectiveness,

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Code and data used in this article are available online at https://github.com/lisiran15/FMHMO.git.

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current hypergraph community detection methods still suffer from two main limitations in terms of functionality and performance.

First, to our knowledge, no current methods can provide fine-grained fuzzy community information for hypergraphs, specifically regarding the partial affiliations among nodes, hyperedges, and hypergraph communities. Although lower- and higher-order fuzzy memberships can describe the partial belongingness of nodes to multiple communities valued continuously within $[0, 1]$, they primarily rely on edges [20], [21] and motifs [22] in ordinary simple graphs, respectively. Thus, they are insufficient to capture the intricate and diverse fuzzy relationships within hypergraph community structures formed by the hyperedges. Furthermore, the identified hypergraph communities are generally characterized by coarse-grained crisp community structures, where each node can only belong to one or more communities in a crisp manner, i.e., the membership grade to each community is either 1 or 0 [4], [23]. Consequently, these crisp communities fail to capture the fine-grained meso-scale community information present in real-world hypergraphs, where a single node (or hyperedge) may have varying degrees of affiliation with multiple communities.

Second, the quality of identified crisp hypergraph communities is significantly influenced by the absence of fine-grained neighbor fuzzy community information of nodes and hyperedges. Specifically, the quality of hypergraph modularity optimization is often limited, primarily due to inaccurate hypergraph reduction [24] and hypergraph partition recovery [11]. Existing hypergraph reduction methods typically simplify the hyperedges by using simpler building blocks like cliques [19]. Consequently, crucial hypergraph topological information, such as node hyperdegrees, can be easily lost or altered [25]. This compromises the quality of communities identified from the reduced simple graph and further limits the quality of the recovered hypergraph partition. Moreover, during the recovery of hypergraph partitions from a partitioned simple graph, existing methods frequently assign nodes to hypergraph communities either randomly or based on coarse-grained discrete topological features (e.g., number of hyperedges involved in communities [23]). Consequently, for hyperedge-based bridge nodes that participate in multiple hyperedges across different communities, the recovered single community affiliation is frequently inaccurate.

To solve the issues, first, we propose novel multi-scale hypergraph fuzzy memberships, which systematically quantify the partial belongingness (i.e., membership grades) among nodes, hyperedges, and hypergraph communities within the continuous range of $[0, 1]$. Thereby, they provide richer fuzzy community information for hypergraphs in a more fine-grained manner. Specifically, the complex structural characteristics of hyperedges, along with their diverse interactions with communities, necessitate a more sophisticated fuzzy membership evaluation than that based on edges [21] or motifs [22]. In particular, not only hyperedge density but also their size, internal node aggregation, and interaction extent with multiple communities should be considered. Therefore, when quantifying node memberships to hypergraph communities, both the memberships of nodes to incident hyperedges and that of these incident hyperedges to communities are measured. These multiscale hypergraph fuzzy memberships facilitate an understanding of fuzzy and overlapping characteristics of nodes and hyperedges within hypergraph community structures.

Second, the multiscale hypergraph fuzzy memberships, employed as fine-grained neighbor fuzzy information, enhance crisp hypergraph community detection via a general framework named fuzzy membership-assisted hypergraph modularity optimization (FMHMO). Specifically, the FMHMO framework includes two key strategies: First, a HS-HR strategy reduces a hypergraph to a simple weighted graph, mapping hyperedges as nodes and their interactions as edges using novel hyperedge similarity as weights. Thereby, preserving intrahyperedge and interhyperedge topologies of the original hypergraph. Second, a fuzzy membership-based hypergraph partition recovery (FM-HPR) strategy recovers a hypergraph partition from the approximately modularity-optimal weighted graph partition. By quantifying the fuzzy memberships of hyperedge-based bridge nodes, we enable the precise assignment of their crisp hypergraph community affiliations. Thereby, enhancing local search capability and improving modularity quality of recovered crisp hypergraph partition. The main contributions of this article are summarized as follows.

1) We define a set of novel concepts for multi-scale hypergraph fuzzy memberships, which systematically quantifies the partial belongingness among nodes, hyperedges, and hypergraph communities within interval [0, 1], providing richer and fine-grained fuzzy community information for hypergraphs.
2) A general framework FMHMO is proposed to approximate the modularity-optimal crisp hypergraph partition, which designs novel hyperedge similarity to preserve sufficient hypergraph topologies in a simple weighted graph, and utilizes fine-grained node fuzzy memberships to enhance accurate hypergraph partition recovery.
3) Experimental results indicate that the proposed FMHMO demonstrates superiority in identifying hypergraph partitions from both synthetic and real-world datasets, outperforming state-of-the-art methods in accuracy and modularity quality.

The rest of this article is organized as follows. Section II briefly reviews related work on hypergraph community detection. Section III details the introduced multi-scale hypergraph fuzzy memberships. Section IV presents the proposed FMHMO framework. In Section V, empirical results to verify the effectiveness of FMHMO are reported. Finally, Section VI concludes this article.

II. Related Work

In recent years, hypergraph community detection has attracted significant attention [4], [12], [26]. Current representative approaches can be roughly classified into three categories: 1) derivative graphs; 2) probabilistic models; and 3) modularity-based optimization.

Derivative graphs: Contreras-Aso et al. [13] introduced Hypergraph-DG, a method for community detection in higher-order hypergraphs. It works by constructing the

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derivative graph of a hypergraph and applying the highest gap cut method to partition it. This approach, in addition to being naturally adapted to higher-order hypergraphs, presents computational advantages over other algorithms used in this context.

Probabilistic models: Contisciani et al. [16] introduced a probabilistic generative model, named Hypergraph-MT, for hypergraphs with mixed-membership community structure. Based on a statistical inference framework, the model provides a principled, efficient and scalable approach to extract overlapping communities in hypergraphs. Hypergraph-MT outperforms clique expansion methods in running time and is suitable for higher-order datasets with large hyperedges. In addition, Ruggeri et al. [4] developed a probabilistic framework to model hypergraphs, including a hypergraph mixed-membership stochastic block model, named Hy-MMSBM. The method allows performing inference on very large hypergraphs, and more accurately detecting both assortative and disassortative community structures.

Modularity-based optimization: Kaminski et al. [9] defined a hypergraph modularity function q_H, generalizing graph modularity by treating hyperedges as indicators of vertex community membership. A CNM-like heuristic algorithm, named SHM-CNM, was developed to find an optimal solution with respect to the hypergraph modularity. Recently, the authors further proposed a scalable hypergraph community detection algorithm with q_H, named h-Louvain [27]. It adjusts the initial stage of the classical Louvain via a dynamically tuned linear combination of the graph modularity function and the hypergraph modularity function. Kumar et al. [11] proposed a generalized modularity maximization framework with an expended hypergraph modularity Q^{hyp}. They then optimized Q^{hyp} using the Louvain method and developed an iterative refinement process for clustering, leveraging higher-order information within the hypergraph to promote balanced hyperedge cuts. In addition, Chodrow et al. [10] provided a highly general hypergraph maximum likelihood Louvain algorithm, named HMLL, for generative hypergraph clustering.

The aforementioned methods have demonstrated effectiveness in discovering crisp hypergraph communities. Particularly, modularity-based optimization has emerged as one of the most promising method, due to the advantages of approximating the global optimum, stability, and few parameters. Despite the success, current detection methods still encounter challenges: none can provide fine-grained hypergraph fuzzy community information, and the quality of identified crisp hypergraph partitions is often limited.

Notably, hypergraph neural networks (HNNs) have gained significant attention in recent years for hypergraph analysis and application. Specifically, Duta et al. [28] introduced two unique formulations of sheaf hypergraph Laplacians (linear and nonlinear), and developed two novel HNN models, namely sheaf hypergraph neural network (SheafHyperGNN) and sheaf hypergraph convolutional network (SheafHyperGCN). Zou et al. [29] proposed a universal feature encoder, termed UniG-Encoder, designed for both graph and hypergraph representation learning. This encoder derives node embeddings by integrating node features and edge/hyperedge topologies through a normalized projection matrix, enabling downstream tasks such as node classification. Gao et al. introduced two HNN frameworks, HGNN [30] and HGNN+ [31], which leverage hyperedge modeling and hypergraph convolution. In particular, the multiple hypergraph convolution layers perform node-edge-node transform, thereby refining node features via hyperedge topologies to extract more discriminative node embeddings. While these HNN methods are primarily designed for the supervised node classification task, the learned node embeddings can also be utilized for unsupervised hypergraph community detection.

III. Multiscale Hypergraph Fuzzy Memberships

A. Challenges of Hypergraph Fuzzy Memberships

Existing fuzzy memberships, such as edge-based lower-order and motif-based higher-order ones, are unsuitable for hypergraphs because of the variable structures of hyperedges and their diverse interactions with communities. Specifically, current fuzzy memberships rely on the connection density of lower-order edges [21], [32], [33] or higher-order motifs [22], which exhibit simpler and fixed structural characteristics. In contrast, hyperedges represent more complex interaction patterns characterized by arbitrary node sizes, variable internal node aggregation, and diversified interactions between hyperedges and communities. Thus, existing fuzzy memberships cannot accurately describe the complex partial community affiliations of nodes in hypergraphs.

To enrich detailed hypergraph fuzzy community information, we introduce novel concepts of multi-scale hypergraph fuzzy memberships that quantify the partial belongingness among nodes, hyperedges, and hypergraph communities within interval $[0, 1]$, by leveraging polyadic complex higher-order connectivity patterns, i.e., hyperedges [14], [15]. For clarity, the formal notations and symbols used in multi-scale hypergraph fuzzy memberships and hypergraph community detection are summarized in Table I.

B. Constructing Multiscale Hypergraph Fuzzy Memberships

Formally, a hypergraph can be represented as $G^H(V^H, E^H)$, where $V^H(|V^H|{=}n)$ denotes the node set and $E^H(|E^H|{=}m)$ indicates the hyperedge set [34]. The initial step establishes a measure of hypergraph node similarity, which is essentially an extension of hyperedge-based Jaccard similarity, quantified with the incident hyperedge topologies of two nodes. Mathematically, the hypergraph node similarity for any two nodes v and v' in G^H is defined as

$$S_{v,v'} = \frac{\left|\Gamma_e^H(v) \cap \Gamma_e^H(v')\right|}{\left|\Gamma_e^H(v) \cup \Gamma_e^H(v')\right|} \tag{1}$$

which equals to the ratio of the intersection to the union of the incident hyperedge sets of the two nodes, where $\Gamma_e^H(v)$ and $\Gamma_e^H(v')$ denotes the set of incident hyperedges of nodes v and v', respectively. Since the value of $S_{v,v'}$ falls within the range $[0, 1]$, it captures the degree of similarity between the hyperedge-based neighborhood topologies of nodes v and v'. An illustration of the hypergraph node similarity is shown in Fig. 1.

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TABLE I
FORMAL NOTATIONS AND SYMBOLS

Notation	Description
G^H	a hypergraph with node set V^H and hyperedge set E^H
$\boldsymbol{C}^H$	a crisp partition of hypergraph G^H, $\boldsymbol{C}^H=\{C_1^H, C_2^H, \cdots, C_c^H\}$
H	incidence matrix of G^H, $H=[h_{ve}]^{n\times m}$, $h_{ve}=\{0,1\}$
A	adjacency matrix of G^H, $A=[a_{ij}]^{n\times m}$
$d^H(v)$	hyperdegree of node v, i.e., number of incident hyperedges
$d(v)$	degree of node v (the number of hyperedge-based neighbor nodes)
$d^H(e)$	degree of hyperedge e, i.e., number of internal nodes within e
$S_{v,v'}$	similarity between nodes v and v' in G^H
$S_{v,e}$	similarity between node v and hyperedge e in G^H
S_{e_a,e_b}	similarity between hyperedges e_a and e_b in G^H
$\Gamma_e^H(v)$	the set of incident hyperedges of node v in G^H
$\Gamma_v^H(v)$	the set of hyperedge-based neighbor nodes of node v in G^H
$\Gamma_C^H(v)$	the set of hyperedge-based neighbor communities of node v in G^H
$u_{v,e}^H$	membership grade of node v to hyperedge e in G^H
$u_{e,C}^H$	membership grade of hyperedge e to community C in G^H
$u_{v,C}^H$	membership grade of node v to community C in G^H
$attr^H(v,C)$	topological attraction of community C to node v in G^H
G^W	a reduced weighted graph with weight matrix $W=[w_{ij}]^{m\times m}$
$\boldsymbol{C}^W$	a crisp partition of the reduced graph G^W
$u_{v_e,C}^W$	membership grade of node v_e to community C in G^W
$attr^W(v_e,C)$	topological attraction of community C to node v_e in G^W
$\Gamma_C^W(v_e)$	the set of edge-based neighbor communities of node v_e in G^W

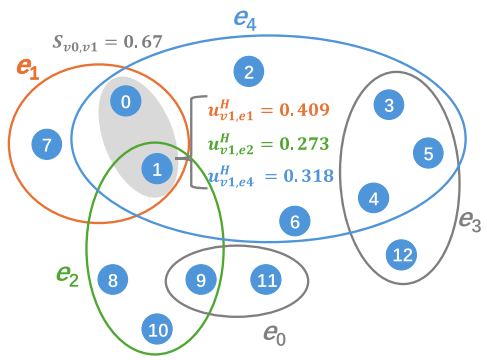


Fig. 1. Illustration of hypergraph node similarity and node fuzzy membership to hyperedges. For two nodes (e.g., v_0 and v_1) in a simple hypergraph, the hypergraph node similarity S_{v_0,v_1} based on incident hyperedge sets, i.e., $\Gamma_e^H(v_0)=\{e_1,e_4\}$ and $\Gamma_e^H(v_1)=\{e_1,e_2,e_4\}$, equals to 0.67. The membership grades for a node (e.g., v_1) to incident hyperedges (e.g., e_1,e_2,e_4), denoted as u_{v_1,e_1}^H, u_{v_1,e_2}^H, and u_{v_1,e_4}^H, are equal to 0.409, 0.273, and 0.318, respectively. The sum of these membership grades is equal to "1".

Using this measure, multiscale hypergraph fuzzy memberships are constructed as follows.

First, the fuzzy memberships of nodes to hyperedges should be considered, due to the variable hyperedge size, internal node aggregation, and participation in multiple hyperedges. Specifically, first, a hyperedge can include any number of nodes, with larger hyperedges having lower individual contributions from each node. Second, topological closeness among nodes in a hyperedge varies, with greater closeness leading to higher participation. Third, a node can belong to multiple hyperedges, reducing its contribution to each.

Formally, for a node v in hypergraph G^H, its membership grade to an incident hyperedge e is defined as the ratio of the similarity between v and e to the total similarity between v and all incident hyperedges in $\Gamma_e^H(v)$.

Definition 1 (Node Membership to Hyperedge): The membership grade of a node v to a hyperedge e is defined as

$$u_{v,e}^H = \frac{S_{v,e}}{\sum_{e_k\in\Gamma_e^H(v)} S_{v,e_k}} = \frac{\sum_{v'\in e,v'\neq v} S_{v,v'}/(d^H(e)-1)}{\sum_{e_k\in\Gamma_e^H(v)}\sum_{v'\in e_k,v'\neq v} S_{v,v'}/(d^H(e_k)-1)} \quad (2)$$

where $S_{v,e}$ represents the similarity between v and e, and $\sum_{e_k\in\Gamma_e^H(v)} S_{v,e_k}$ indicates the total similarity between v and all incident hyperedges. In particular, $S_{v,e}$ is defined as the average node similarity between v and all nodes within e except for node v, i.e., $\sum_{v'\in e,v'\neq v} S_{v,v'}/(d^H(e)-1)$. Here, $S_{v,v'}$ represents the hypergraph node similarity between v and each other node v' within hyperedge e that is different from node v, i.e., $v'\neq v$. This helps to emphasize the variation in topological attractiveness between node v and multiple incident hyperedges, thereby quantifying the diverse similarity between node v and each incident hyperedge e more discriminately. In addition, $d^H(e)$ denotes the degree of hyperedge e, which equals to the total number of internal nodes in e. The value of $u_{v,e}^H$ ranges from 0 to 1, with larger values indicating a higher average node similarity between v and e, thereby implying a stronger membership grade. The sum of membership grades of node v to all incident hyperedges is equal to "1". An illustration of node fuzzy memberships to incident hyperedges are shown in Fig. 1.

Second, the fuzzy memberships of hyperedges to communities are considered, owning to the variable hyperedge size, internal node aggregation, and interaction with multiple communities. Formally, for a hyperedge e in hypergraph G^H, its membership grade to any neighbor community C is defined as the proportion of nodes in e belonging to C. Here, the neighbor community of e refers to a hypergraph community that encompasses any internal node within e.

Definition 2 (Hyperedge Membership to Community): The membership grade of a hyperedge e to a neighbor hypergraph community C is defined as

$$u_{e,C}^H = \frac{|e\wedge C|}{d^H(e)} \quad (3)$$

where $|e\wedge C|$ represents the number of nodes that co-occur in e and C. The value of $u_{e,C}^H$ ranges from 0 to 1, with larger values indicating a higher proportion of shared nodes, thereby implying a stronger membership grade of e to C. The sum of membership grades of e to all neighbor communities is equal to "1". An illustration of hyperedge fuzzy memberships to neighbor communities are shown in Fig. 2 .

At last, the node fuzzy memberships to communities based on hyperedges are measured, considering complex factors such as connection density, hyperedge size, internal node aggregation, and interaction with multiple communities. An example is illustrated in Fig. 3, where node v_0 exhibits a higher membership grade to the neighbor community C_2 due to the denser connections of edges and motifs depicted in Fig. 3(a) and (b), respectively. Conversely, despite the less dense hyperedge connections, node v_0 should have a larger membership

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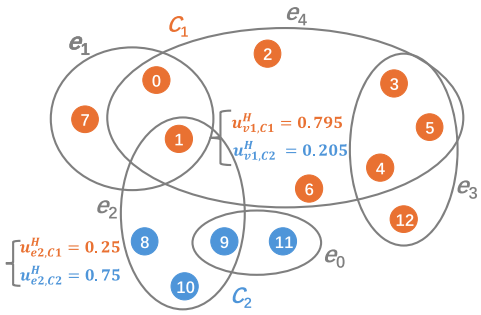


Fig. 2. Illustration of fuzzy memberships for hyperedges and nodes to communities. The membership grades of a hyperedge (e.g., e_2) to neighbor communities (e.g., C_1 and C_2), denoted as $u^H_{e_2,C_1}$ and $u^H_{e_2,C_2}$, are 0.25 and 0.75, respectively, with a total sum of "1". The membership grades of a node (e.g., v_1) to neighbor communities (e.g., C_1 and C_2), denoted as $u^H_{v_1,C_1}$ and $u^H_{v_1,C_2}$, are 0.795 and 0.205 respectively, with a total sum of "1".

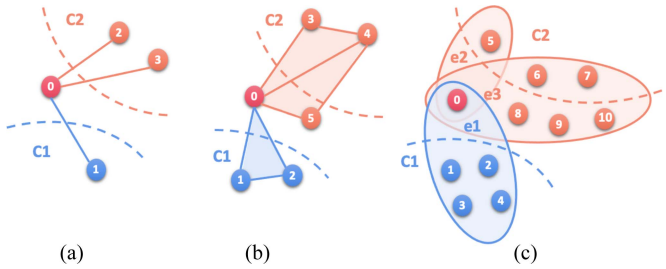


Fig. 3. Neighborhood topologies of a node v_0 to two neighbor communities (i.e., C_1 and C_2) based on (a) lower-order edges, (b) simple higher-order triangle motifs, and (c) polyadic complex higher-order hyperedges.

grade to C_1 because of stronger interaction strength (4 common nodes) compared to C_2 (3 common nodes), as shown in Fig. 3(c).

Formally, for a node v in hypergraph G^H, its fuzzy membership to a neighbor community C is quantified as the ratio of the attraction of C to v to the sum of attractions of all neighbor communities to v. In particular, the attraction of a specific community C depends on the product of the membership grades of node v to incident hyperedges and that of the incident hyperedges to community C.

Definition 3 (Node Membership to Community): The membership grade of a node v to a hyperedge-based neighbor community C can be expressed as

$$u^H_{v,C} = \frac{attr^H(v,C)}{\sum_{C_k \in \Gamma^H_C(v)} attr^H(v,C_k)} \tag{4}$$

where $\Gamma^H_C(v)$ is the set of hyperedge-based neighbor communities, i.e., all hypergraph communities containing any hyperedge-based neighbor node of v. Here, the set of hyperedge-based neighbor nodes of v is represented as $\Gamma^H_v(v)$, that is, the collection of nodes that sharing any hyperedge with v. In particular, $attr^H(v,C_k)$ represents the topological attraction of any neighbor community C_k to node v, i.e.,

$$attr^H(v,C_k) = \sum_{e \in \Gamma^H_e(v)} u^H_{v,e} \cdot u^H_{e,C_k} \tag{5}$$

where $\Gamma^H_e(v)$ is the set of incident hyperedges of node v, and $u^H_{v,e}$ and u^H_{e,C_k} represent the membership grade of node v to any incident hyperedge e and that of hyperedge e to any neighbor community C_k, respectively. The value of $u^H_{v,C}$ ranges from 0 to 1, with larger values indicating a stronger topological interaction between v (with incident hyperedges) and C, thereby implying a stronger membership grade. The sum of membership grades of v to all neighbor communities is equal to "1". An illustration of node fuzzy memberships to neighbor communities are shown in Fig. 2.

C. Application of Hypergraph Fuzzy Memberships

The multiscale hypergraph fuzzy memberships can be applied to systematically quantify fuzzy relations in real-world hypergraphs, providing richer fuzzy community information and enhancing the accuracy and modularity quality of identified crisp communities. For clarity, an example of a real-world social hypergraph, namely Southern Women hypergraph, consisting of 18 nodes (women), 14 hyperedges (social events), and two ground-truth crisp communities C_1 (depicted by orange nodes) and C_2 (represented by green nodes), is illustrated in Fig. 4(a). In addition, the fuzzy membership of each hyperedge and node to the two identified communities is presented in Fig. 4(b) and (c), respectively.

First, the multiscale hypergraph fuzzy memberships accurately reveal the fine-grained partial affiliations for all 14 hyperedges and 18 nodes to the two ground-truth communities. Specifically, as shown in Fig. 4(b), hyperedges e_1-e_5 have full membership with C_1, and e_{10}-e_{14} have full membership with C_2. Hyperedges e_6, e_7, e_8 and e_9 exhibit varying membership grades within the interval $[0,1]$, with e_7 and e_8 showing the most ambiguity. In addition, as depicted in Fig. 4(c), the 18 nodes have partial memberships to C_1 and C_2 within $[0,1]$, with v_7 and v_{15} showing the most ambiguous membership grades. These fuzzy memberships for all hyperedges and nodes align accurately with the ground-truth crisp hypergraph community structure, providing reliable fuzzy community information for the real-world hypergraph.

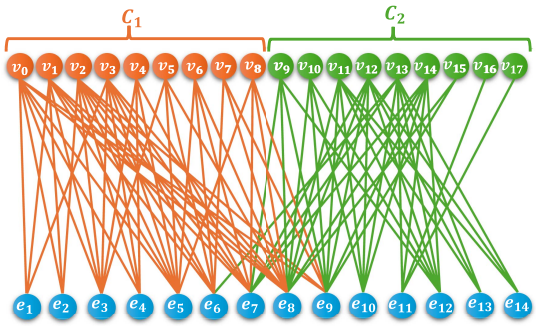
Second, the use of multiscale hypergraph fuzzy memberships improves the accuracy and modularity of crisp hypergraph communities. Specifically, fuzzy memberships of hyperedges and nodes provide useful neighbor fuzzy information for optimizing hypergraph modularity within the proposed FMHMO framework (Section IV-C). For the real-world Southern Women hypergraph depicted in Fig. 4(a), FMHMO with multi-scale fuzzy memberships achieves optimal accuracy (NMI [35], ARI [36], F_1 Score [37], and FMI [38] all equal to 1) and hypergraph modularity (Q^{hyp} [11] equals 0.37). These results outperform four state-of-the-art hypergraph community detection methods: Hypergraph-DG [13], Hypergraph-MT [16], Hy-MMSBM [4], and SHM-CNM [9].

IV. Modularity-Based Hypergraph Community Detection With Fuzzy Memberships

A. Definition of Hypergraph Community Detection

Formally, for a hypergraph $G^H(V^H, E^H)$ with node set $V^H(|V^H|=n)$ and hyperedge set $E^H(|E^H|=m)$, an incidence

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(a)

Hyperedges	u^H_{e,C_1}	u^H_{e,C_2}	Crisp Partition
e_1	1.00	0.00	(1,0)
...	...	...	...
e_5	1.00	0.00	(1,0)
e_6	0.88	0.12	(1,0)
e_7	0.40	0.60	(0,1)
e_8	0.43	0.57	(0,1)
e_9	0.33	0.67	(0,1)
e_{10}	0.00	1.00	(0,1)
...	...	...	...
e_{14}	0.00	1.00	(0,1)

(b)

Nodes	u^H_{v,C_1}	u^H_{v,C_2}	Crisp Partition
v_0	0.87	0.13	(1,0)
v_1	0.86	0.14	(1,0)
v_2	0.81	0.19	(1,0)
v_3	0.87	0.13	(1,0)
v_4	0.91	0.09	(1,0)
v_5	0.87	0.13	(1,0)
v_6	0.75	0.25	(1,0)
v_7	0.54	0.46	(1,0)
v_8	0.59	0.41	(1,0)
v_9	0.40	0.60	(0,1)
v_{10}	0.22	0.78	(0,1)
v_{11}	0.17	0.83	(0,1)
v_{12}	0.25	0.75	(0,1)
v_{13}	0.23	0.77	(0,1)
v_{14}	0.24	0.76	(0,1)
v_{15}	0.48	0.52	(0,1)
v_{16}	0.17	0.83	(0,1)
v_{17}	0.17	0.83	(0,1)

(c)

Fig. 4. Applying multiscale hypergraph fuzzy memberships on a real-world Southern Women hypergraph. (a) The topological structure of the hypergraph, comprising 18 nodes (women),14 hyperedges (social events), and two ground-truth crisp communities: C_1 (orange nodes) and C_2 (green nodes). (b) Fuzzy memberships of all 14 hyperedges to the two crisp communities. (c) Fuzzy memberships of all 18 nodes to the two ground-truth crisp communities.

matrix $H = [h_{ve}]$ can be defined, where $h_{ve} = 1$ if node v belongs to the hyperedge e; otherwise, $h_{ve} = 0$. On this basis, an adjacency matrix, i.e., $A = [a_{ij}]^{n\times m}$, can be derived as

$$a_{ij} = \left[HH^T - D\right]_{ij} \tag{6}$$

where a_{ij} denotes the number of hyperedges containing nodes v_i and v_j. D is a diagonal matrix, where D_{ii} represents the number of hyperedges node v_i belongs to, also known as the hyperdegree of v_i, i.e., $d^H(v_i)$. Additionally, the degree of v_i is denoted as the number of hyperedge-based neighbor nodes, that is, $d(v_i) = \sum_{j=1}^{n} \widetilde{a}_{ij}$. $\widetilde{A} = [\widetilde{a}_{ij}]^{n\times m}$ is the binarized adjacency matrix of A, where element $\widetilde{a}_{ij} = 1$ if v_i and v_j share at least one hyperedge, and $\widetilde{a}_{ij} = 0$ otherwise. In detail, it can be defined as

$$\widetilde{a}_{ij} = \begin{cases} 1, & \text{if } a_{ij} > 0, \\ 0, & \text{if } a_{ij} = 0. \end{cases} \tag{7}$$

Furthermore, a hyperedge can connect multiple nodes. The number of nodes in a hyperedge e is defined as its degree, represented as $d^H(e) = \sum_{i=1}^{n} h_{ie}$ [23].

Hypergraph community detection extends graph-based methods by capturing polyadic complex group interactions beyond pairwise relationships [12]. Essentially, it involves finding a hypergraph partition, where hyperedge connections of nodes within each community are expected to be dense, whereas those between communities should be sparse [13]. Given a hypergraph $G^H(V^H, E^H)$, a crisp hypergraph partition $\boldsymbol{C}^H = \{C_1^H, C_2^H, \ldots, C_c^H\}$ divides nodes into a family of groups, such that the node density and hyperedge-based interaction strength within the same group are relatively large, while that between different groups are relatively small [39].

In particular, hypergraph community detection based on hyperedges poses significantly greater challenges than traditional lower-order (edge-based) and higher-order (motif-based) community detection. Unlike these methods, which focus on edge or motif density, hypergraph community detection must also account for variable hyperedge topologies and intricate multiscale topological interaction strengths. Therefore, hypergraph community detection has become one of the most challenging problems in the field.

B. Hypergraph Modularity Optimization

A modularity-based hypergraph community detection model is developed, which leverages an approximated modularity-optimal partition of a reduced ordinary weighted graph to achieve high-quality hypergraph partition. In particular, a hypergraph partition seeks to maximize intracommunity node density and hyperedge-based interaction strength while minimizing intercommunity interactions [12]. To this end, we first simplify the hypergraph into a weighted graph while preserving the crucial higher-order hyperedge topologies. Then, we partition the graph by optimizing weighted modularity to maximize intracommunity weights and minimize inter-community weights. Finally, we recover the approximate modularity-optimal graph partition to obtain the community structure of the original hypergraph.

The weighted modularity optimization of the reduced weighted graph G^W, is defined as a global maximization problem, i.e.,

$$F(\boldsymbol{C}^{W*}) = \max(F(\boldsymbol{C}^W)) \tag{8}$$

where $\boldsymbol{C}^W$ stands for a candidate partition of G^W in the search space $\boldsymbol{\Omega}$. $F : \boldsymbol{\Omega} \rightarrow \boldsymbol{R}$ represents an objective function used to evaluate the weighted graph partitions. Here, the weighted modularity Q_W [40] is employed as the objective function, which is

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defined as

$$Q_W = \frac{1}{2\|W\|}\sum_{ij}\left(w_{ij} - \frac{s_i s_j}{2\|W\|}\right)\delta(i,j) \tag{9}$$

where $\|W\|=\frac{1}{2}\sum_{ij} w_{ij}$ is the sum of edge weights in G^W, s_i and s_j represent the strengths of v_i and v_j, equaling to the sum of weights $\sum_{j\in N_i} w_{ij}$ and $\sum_{i\in N_j} w_{ij}$, respectively. N_i and N_j denote the edge-based neighbor sets of v_i and v_j, respectively. $\delta(i,j)$ yields one if nodes v_i and v_j belong to the same community, zero otherwise. The partition with the highest Q_W value is output as the approximate modularity-optimal weighted graph partition, denoted as $\boldsymbol{C}^{W*}$.

C. General Framework of FMHMO

A novel framework, named FMHMO is proposed for hypergraph community detection. The core idea is to incorporate the multi-scale hypergraph fuzzy memberships as fine-grained neighbor fuzzy information to enhance hypergraph modularity optimization, improving the accuracy and modularity quality of the identified hypergraph partition.

The general framework of FMHMO is outlined in Algorithm 1, encompassing hypergraph reduction, weighted modularity optimization, and hypergraph partition recovery. In particular, FMHMO does not employ any iterative approach in order to achieve a balance between modularity quality and time efficiency. Specifically, the three steps are formally delineated as follows.

1) *Hypergraph Reduction:* Reduce the original hypergraph G^H into a simple weighted graph G^W using a HS-HR strategy. In particular, HS-HR maps entire hyperedges in G^H to nodes in G^W, and represents the interactions between hyperedges in G^H as pairwise edges in G^W. Specifically, a hyperedge similarity measure is developed to quantify the edge weights, capturing the interaction strength between hyperedges at both node and local topology levels. HS-HR preserves sufficient original hypergraph topological information in G^W, providing a solid foundation for obtaining higher-quality hypergraph partitions.
2) *Weighted Modularity Optimization:* Generate an approximate modularity-optimal partition $\boldsymbol{C}^{W*}$ of the reduced weighted graph G^W using a universal baseline modularity-based community detection method (e.g., Louvain [41], [42] and Leiden [43]) with the weighted modularity Q_W. This step seeks to achieve rapid and broadly applicable community detection on G^W, thereby emphasizing simplicity and computational efficiency.
3) *Hypergraph Partition Recovery:* Recover the hypergraph partition $\boldsymbol{C}^H$ from the approximate optimal partition $\boldsymbol{C}^{W*}$ using the FM-HPR strategy. In particular, fine-grained fuzzy memberships of hyperedge-based bridge nodes are quantified to enable precise assignment to crisp hypergraph communities of hyperedges in $\boldsymbol{C}^{W*}$. This improves node local search capabilities and facilitates quality improvements of identified crisp hypergraph partition $\boldsymbol{C}^H$, as measured by the hypergraph modularity Q^{hyp}.

Algorithm 1: General Framework of FMHMO.

Input: G^H: original hypergraph; $\boldsymbol{A}$: adjacency matrix;
Output: $\boldsymbol{C}^H$: crisp hypergraph partition of nodes;
1: $G^W \leftarrow$ reduce hypergraph G^H into a simple weighted graph G^W by HS-HR strategy;
2: $\boldsymbol{C}^{W*} \leftarrow$ identify the approximate modularity-optimal weighted graph partition $\boldsymbol{C}^{W*}$ of G^W using Q_W and a universal baseline modularity optimization method;
3: $\boldsymbol{C}^H \leftarrow$ quantify node fuzzy memberships and recover the crisp hypergraph partition $\boldsymbol{C}^H$ of nodes from $\boldsymbol{C}^{W*}$ by FM-HPR strategy;
4: **return** $\boldsymbol{C}^H$

In particular, the performance of the FMHMO framework primarily rely on the two innovative strategies of HS-HR and FM-HPR. In addition, in the FMHMO framework, novel assistance topologies are developed, encompassing: 1) hyperedge-based neighbor nodes and communities; 2) multiscale similarity metrics, including node-to-node, node-to-hyperedge, and hyperedge-to-hyperedge; and 3) multiscale hypergraph fuzzy memberships. These innovative assistance topologies are integrated into the hypergraph modularity optimization, thereby enhancing accuracy and modularity quality of identified hypergraph partitions.

D. Hypergraph Reduction Based on Hyperedge Similarity

Hypergraph reduction is commonly employed to reduce an original hypergraph into a simplified weighted graph with pairwise relationships [24]. The performance of hypergraph reduction is crucial for the quality of weighted graph partition, thereby influencing the accuracy and modularity quality of the recovered hypergraph partition. However, existing methods often oversimplify polyadic complex group interactions among nodes based on hyperedges by utilizing simpler building blocks such as cliques [11], [19]. These simpler building blocks often fail to adequately capture the important higher-order hyperedge topologies in the hypergraph. Consequently, the quality of both the identified simple graph partition and the recovered hypergraph partition is compromised.

To address this issue, in the FMHMO framework, a HS-HR strategy is proposed. The core idea is to maintain the intrahyperedge and interhyperedge topologies through a dual simple weighted graph. Specifically, HS-HR maps entire hyperedges in G^H to nodes in G^W, thereby preserving comprehensive intrahyperedge topologies. In addition, HS-HR maps the interactions between hyperedges in G^H as pairwise edges in G^W. A novel measure of hyperedge similarity is introduced to capture the diverse interaction strengths between hyperedges, which are subsequently used as edge weights in G^W. Consequently, HS-HR ensures that the initial hypergraph topologies regarding the higher-order hyperedges, such as node degree sequences, node hyperdegrees, and intrahyperedge node interactions [25], are comprehensively preserved to the greatest extent possible.

The novel measure of hyperedge similarity considers two factors: 1) hypergraph node similarity among all nodes and 2)

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hyperedge structural similarity of common nodes [23]. Specifically, for the first factor, the attractions among nodes primarily consider the average hypergraph node similarity for all pairs of nodes, focusing on the hyperedge-based topological neighborhoods (i.e., incident hyperedge sets) of any two nodes. For the second factor, hyperedge structural similarity mainly considers the structural participation ratio of common nodes in the two specific hyperedges, relative to their participation in the hyperedge-based topological neighborhoods of all intersection and union nodes. The stronger the two aspects, the higher the similarity between the two hyperedges.

Formally, for a hypergraph G^H, the similarity between any two hyperedges e_a and e_b is defined as

$$S_{e_a e_b} = \gamma \frac{\sum_{v_i \in e_a, v_j \in e_b} S_{v_i, v_j}}{d^H(e_a)\, d^H(e_b)} + (1-\gamma) \frac{2\,|e_a \cap e_b|}{\sum_{v_i \in e_a \cap e_b} d^H(v_i)} \cdot \frac{2\,|e_a \cap e_b|}{\sum_{v_j \in e_a \cup e_b} d^H(v_j)} \tag{10}$$

where the first component indicates the average attraction of all pairs of nodes in e_a and e_b, S_{v_i,v_j} represents the hypergraph node similarity of nodes v_i and v_j in terms of their hyperedge neighborhoods, and $d^H(e_a)$ and $d^H(e_b)$ indicate the degree of e_a and e_b, respectively. The second component represents the hyperedge structural similarity of common nodes (i.e., $|e_a \cap e_b|$). Particularly, $\frac{2|e_a \cap e_b|}{\sum_{v_i \in e_a \cap e_b} d^H(v_i)}$ represents the ratio of structural participation of common nodes in e_a and e_b relative to their participation in all incident hyperedges of intersection nodes. $\frac{2|e_a \cap e_b|}{\sum_{v_j \in e_a \cup e_b} d^H(v_j)}$ represents the ratio of structural participation of common nodes in e_a and e_b relative to their participation in all incident hyperedges of union nodes. A control factor γ adjusts the impact of the two components. The hyperedge similarity $S_{e_a e_b}$ ranges from 0 to 1, and a higher value indicates stronger similarity between e_a and e_b.

Based on the hyperedge similarity, an original hypergraph G^H can be reduced to a simple weighted graph G^W. A weighted adjacency matrix $\boldsymbol{W}=[w_{ij}]^{m\times m}$ can be constructed, where m is the number of hyperedges in G^H. Each element w_{ij} represents the hyperedge similarity between any two nodes v_{e_i} and v_{e_j} in G^W (i.e., hyperedges e_i and e_j in G^H), that is, $w_{ij} = S_{e_i e_j}$. For clarity, an illustration of the HS-HR strategy is shown in Fig. 5, where a hypergraph with five hyperedges is reduced to an ordinary weighted graph with five nodes.

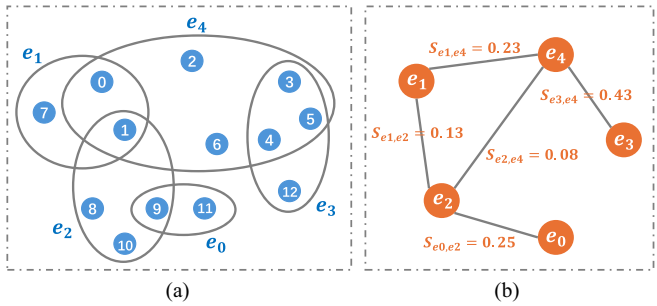


Fig. 5. Illustration of the hyperedge-similarity-based hypergraph reduction (HS-HR) strategy. (a) A simple hypergraph comprising 13 nodes and 5 hyperedges. (b) A reduced weighted graph, where the five nodes represent the original hyperedges, and the 5 edges reflect the interactions between the original hyperedges, with their similarities serving as edge weights.

E. Fuzzy Membership-Based Hypergraph Partition Recovery

Hypergraph partition recovery is crucial in recovering the hypergraph communities of nodes from the weighted graph partition [11]. Particularly, the crisp community assignment for a hyperedge-based bridge node is usually challenging, because the two or more incident hyperedges involved may belong to different communities. Current methods, lacking fine-grained neighbor fuzzy community information, often use random assignment or rely on coarse-grained discrete topology features (e.g., number of hyperedges [23]). This may result in unreasonable allocation of these bridge nodes, thereby compromising the accuracy of hypergraph partition recovery.

To address this issue, in the FMHMO framework, a fuzzy-membership-based hypergraph partition recovery (FM-HPR) strategy is proposed. Specifically, given the weighted graph partition (i.e., hypergraph partition at the level of hyperedges), the crisp community assignment for each non-bridge node can be directly determined by its single incident hyperedge. To accurately determine the crisp community assignment for the remaining bridge nodes, FM-HPR first quantifies the fuzzy membership of each bridge node to the crisp hypergraph communities associated with multiple incident hyperedges. These detailed fuzzy memberships effectively enhance the local search capability for each bridge node within neighborhood. Consequently, each bridge node is accurately assigned to the community with the highest membership grade (i.e., strongest hyperedge-based topological interactions), thereby facilitating the recovery of crisp hypergraph partition of nodes with high modularity quality with respect to Q^{hyp} [11]. At the same time, the hypergraph fuzzy community information can be obtained, mainly including node fuzzy memberships to multiple neighbor hypergraph communities.

The FM-HPR strategy comprises three primary steps: 1) selecting objective bridge nodes; 2) calculating node fuzzy memberships to neighbor communities; and 3) determining community affiliation. For clarity, an example of the recovery process is illustrated in Fig. 6. First, a set of objective hyperedge-based bridge nodes, i.e., V_{obj}, is selected from the original hypergraph G^H. Each objective node has multiple incident hyperedges that involve distinct crisp hypergraph communities of hyperedges. For example, a bridge node v_1 in Fig. 6(b) has three incident hyperedges (i.e., e_1, e_2, and e_4). Because these hyperedges are divided into two different communities (i.e., C_1 and C_2) in the graph partition $\boldsymbol{C}^{W*}$, it is difficult to determine the crisp community affiliation of v_1 in the hypergraph partition $\boldsymbol{C}^H$.

Second, for each objective bridge node v in V_{obj}, we compute its membership grade to each hyperedge-based neighbor community using the formula presented in (4). Particularly, the component $u^H_{e,C}$ denotes the membership grade of an incident hyperedge e of node v to a specific

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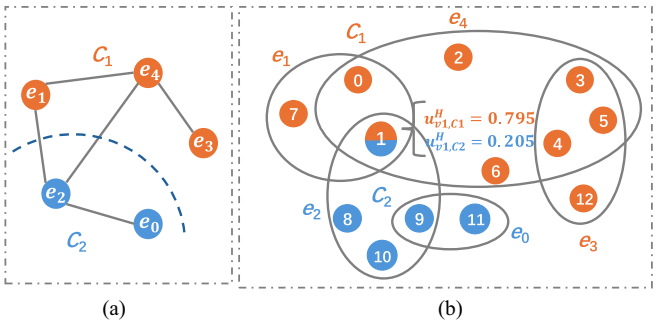


Fig. 6. Illustration of the fuzzy membership-based hypergraph partition recovery (FM-HPR) strategy. (a) Identified partition of the reduced simple weighted network G^W, including two communities C_1 and C_2. (b) Determining community affiliation of a bridge node v_1 in the hypergraph G^H based on its highest membership grade to the two neighbor communities C_1 and C_2.

hyperedge-based neighbor community C, which is calculated based on the identified weighted graph partition $\boldsymbol{C}^{W*}$, as illustrated in Fig. 6(a). Notably, since the hyperedges in G^H are mapped as nodes in G^W, $u^H_{e,C}$ is equivalent to the membership grade $u^W_{v_e,C}$, representing the partial affiliation of the node v_e in G^W (i.e., incident hyperedge e of node v in G^H) to community C. For instance, node v_1 in G^H has three incident hyperedges $\{e_1, e_2, e_4\}$ as shown in Fig. 6(b), corresponding to three nodes in G^W depicted in Fig. 6(a). Thus, the membership grades $u^H_{e_1,C}$, $u^H_{e_2,C}$, and $u^H_{e_4,C}$ can be equated to $u^W_{v_{e1},C}$, $u^W_{v_{e2},C}$ and $u^W_{v_{e4},C}$, respectively.

Definition 4 (Hyperedge-based Node Membership to Community): The membership grade $u^W_{v_e,C}$ of a hyperedge-based node v_e in the weighted graph G^W to a specific neighbor community C can be expressed as

$$u^W_{v_e,C} = \frac{attr^W(v_e, C)}{\sum_{C_k \in \mathcal{T}^W_C(v_e)} attr^W(v_e, C_k)} \tag{11}$$

where $\mathcal{T}^W_C(v_e)$ indicates the set of edge-based neighbor communities of node v_e, i.e., all communities in $\boldsymbol{C}^{W*}$ including any edge-based neighbor node of v_e. $attr^W(v_e, C_k)$ represents the topological attractiveness of any edge-based neighbor community C_k to node v_e, i.e.,

$$attr^W(v_e, C_k) = \frac{\sum_{v'_e \in C_k} w_{v_e,v'_e}}{avg_{dis}(v_e, C_k)} \tag{12}$$

where w_{v_e,v'_e} is the weight (i.e., hyperedge similarity) between node v_e and any node v'_e in community C_k. $avg_{dis}(v_e, C_k)$ represents the average shortest path length between node v_e and all nodes in C_k, computed using the *Dijkstra* algorithm [44] based on the topological distances. Based on the concepts, as shown in Fig. 6(b), the membership grades of node v_1 to the two neighbor communities (i.e., C_1 and C_2), denoted as $u^H_{v_1,C_1}$ and $u^H_{v_1,C_2}$, can be calculated as 0.795 and 0.205, respectively.

At last, we assign each objective bridge node to the neighbor community with the highest membership grade. This enhances the probability that each bridge node will be assigned to the community with the strongest hyperedge topological interaction. For instance, node v_1 is assigned to the neighbor community C_1, because v_1 has a higher membership grade to C_1 than that to C_2. Therefore, the local search capabilities of objective bridge nodes can be effectively improved, promoting the quality in recovering the hypergraph partition.

F. Time Complexity Analysis

The computation of the FMHMO framework involves three main components: 1) the HS-HR strategy; 2) a baseline weighted modularity optimization method; and 3) the FM-HPR strategy. Given a hypergraph G^H with n nodes and m hyperedges, first, reducing the G^H to a simple weighted graph G^W using the HS-HR strategy costs $O(|E^W| \cdot \langle d^H(e)\rangle^2 \cdot \langle d^H(v)\rangle^2)$, where $|E^W|$ represents the average number of pair-wise edges in G^W, $\langle d^H(e)\rangle$ represents the average degree of hyperedges in G^H, and $\langle d^H(v)\rangle$ represents the average hyperdegree of nodes in G^H. Second, the weighted modularity optimization using a typical baseline method (e.g., Louvain [41], [42] and Leiden [43]) in G^W generally costs $O(m \cdot \log(m))$, and the evaluation of the weighted modularity Q_W incurs a computational cost of $O(m^2)$. Third, recovering a hypergraph partition $\boldsymbol{C}^H$ with the FM-HPR strategy cost $O(|V_{obj}| \cdot c \cdot \langle d^H(e)\rangle \cdot \langle d^H(v)\rangle^2)$, where $|V_{obj}|$ represents the number of objective bridge nodes, c is the number of communities in $\boldsymbol{C}^{W*}$. Evaluating $\boldsymbol{C}^H$ using the hypergraph modularity Q^{hpy} incurs a computational cost of $O(n^2)$ [11].

In summary, the time complexity of FMHMO can be expressed as $O(|E^W| \cdot \langle d^H(e)\rangle^2 \cdot \langle d^H(v)\rangle^2)$+$O(m \cdot \log(m))$+$O(m^2)$+$O(|V_{obj}| \cdot c \cdot \langle d^H(e)\rangle \cdot \langle d^H(v)\rangle^2)$+$O(n^2)$, which can be simplified to $O(m^2 + n^2)$. Compared to the number of nodes n and hyperedges m, the other components can be considered as constants due to their significantly smaller values. The time complexity of FMHMO is relatively low compared to typical modularity-based hypergraph community detection methods [9], [11].

V. Empirical Results

A. Experimental Settings

1) Comparison Algorithms: The performance of FMHMO is compared with state-of-the-art hypergraph community detection algorithms, namely, Hypergraph-DG [13], Hypergraph-MT [16], Hy-MMSBM [4], and SHM-CNM [9]. In particular, a supervised HNN-based hypergraph community detection algorithm, named HNNCD, is developed for performance comparison. Specifically, HNNCD employs a supervised two-layer HGNN model comprising two hyperedge convolutional layers [30], [31], which performs node-edge-node transform, thereby refining node feature representations regarding meso-scale hypergraph communities by leveraging the hypergraph topologies. Both node features and node labels are manually constructed to encode meso-scale hypergraph community structural information. Node labels are assigned based on the ground-truth community affiliation of each node. By training the HGNN model, refined node embeddings are obtained, which are then clustered using the classical k-means algorithm to identify hypergraph communities.

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TABLE II
STATISTICS OF REAL-WORLD HYPERGRAPHS USED IN THE EXPERIMENTS

Network	n	m	$d^H_{max}(e)$	c
Hospital	75	1,825	5	4
Workspace	92	788	4	5
PrimarySchool	242	12,704	5	10
HighSchool	327	7,818	5	17
HouseCommittees	1,290	335	81	2
Cora_ca	2,708	1,072	5	7
Citeseer	3,312	1,079	26	6
EnronEmail	4,423	5,734	25	2
GenDisease	9,262	3,128	1,074	2
Pubmed	19,717	7,531	171	3

n represents the number of nodes, m represents the number of hyperedges, $d^H_{max}(e)$ represents the maximum hyperedge degree, and c is the number of real communities.

The FMHMO algorithm exhibits low sensitivity to parameter settings. The primary parameter, the control factor γ in the HS-GR strategy, is empirically assigned a value of either 0 or 0.5 for real-world hypergraphs and a fixed value of 0.5 for synthetic H-ABCD benchmark hypergraphs. The parameters for the four competitors are set according to the recommendations [4], [9], [13], [16]. For HNNCD, the clustering parameter k is initially set within the range $[2, 2c]$, where c represents the number of ground-truth communities, and the optimal value of k is determined by applying the silhouette coefficient method.

2) Real-World Networks: The algorithms are tested on ten typical real-world hypergraphs with different types, scales, and characteristics. Table II lists the detailed information of the hypergraphs, which can be consulted from the open-source hypergraph datas [45]. Specifically, more than 60% of the hypergraphs have more than 1000 nodes, where the Pubmed network has the largest scale, containing 19 717 nodes and 7531 hyperedges.

3) Synthetic Benchmark Networks: The widely used synthetic benchmark hypergraphs, i.e., Hypergraph Artificial Benchmark for Community Detection (H-ABCD) [46], are adopted for performance evaluation. In H-ABCD hypergraphs, the noise parameter ξ controls the fraction of non-community hyperedges. The average and maximum node degree are set to 5 and 30, and minimum and maximum community size are set to 20 and 100, respectively. The ground-truth partition of each H-ABCD hypergraph can be automatically obtained according to predefined parameters during construction.

4) Evaluation Metrics: Five performance metrics are utilized to evaluate the crisp hypergraph partitions of nodes identified by the FMHMO framework and state-of-the-art hypergraph community detection approaches. Specifically, four classification accuracy metrics, including Normalized Mutual Information (NMI) [35], Adjusted Rand Index (ARI) [36], F_1 Score [37], and Fowlkes–Mallows Index (FMI) [38], are incorporated to assess the accuracy of the identified crisp hypergraph partitions by comparing them to the ground-truth crisp communities of both synthetic H-ABCD benchmarks and real-world hypergraphs. In particular, the standard NMI measure has been found to exhibit a significant bias as the number of communities increases [47], making it unsuitable as a sole metric for accuracy evaluation. By integrating three additional representative accuracy metrics, i.e., ARI, F_1, and FMI, a more objective and comprehensive accuracy evaluation can be achieved. In addition, one quality metric employs the well-known hypergraph modularity Q^{hyp} [11], which evaluates the significance of hypergraph community structures in terms of node aggregation and module properties without requiring ground-truth information. For all five metrics, NMI, ARI, F_1, and FMI range from $[0, 1]$ and Q^{hyp} ranges from $[-1, 1]$, higher values indicate better detection outcomes.

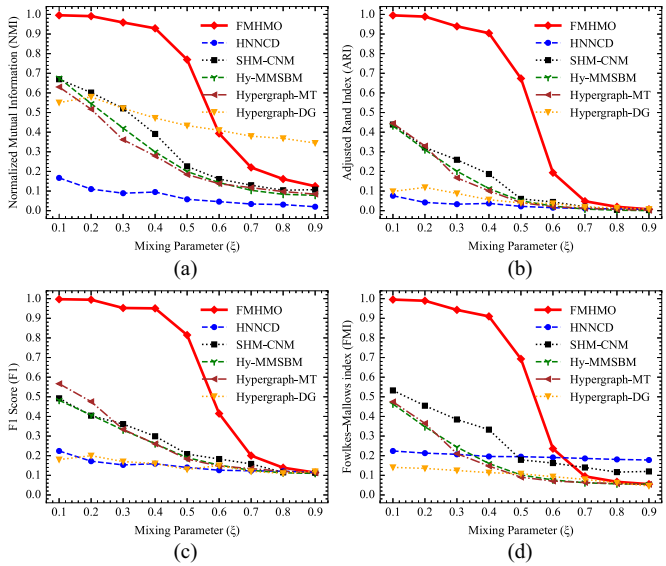


Fig. 7. Average values of four accuracy metrics, i.e., NMI, ARI, F_1, and FMI, derived from 20 independent runs of FMHMO and five state-of-the-art hypergraph community detection methods over a group of synthetic H-ABCD benchmarks with noise parameter ξ increased from 0.1 to 0.9. (a) The average NMI values. (b) The average ARI values. (c) The average F_1 values. (d) The average FMI values.

B. Performance on Synthetic Benchmarks

In this section, first, we assess the performance of FMHMO on synthetic benchmark hypergraphs. The performance of FMHMO is benchmarked against five state-of-the-art hypergraph community detection algorithms: Hypergraph-DG [13], Hypergraph-MT [16], Hy-MMSBM [4], SHM-CNM [9], and HNNCD. A group of nine synthetic H-ABCD benchmarks with a scale of 1000 nodes are adopted, and the noise parameter ξ ranges from 0.1 to 0.9 with interval 0.1.

Fig. 7 presents the average values of four accuracy metrics, i.e., NMI, ARI, F_1 and FMI, obtained by FMHMO and five competitors in 20 independent runs on the H-ABCD hypergraph set. It can be found that FMHMO performs the best in terms of the four accuracy metrics in most test cases, demonstrating significant superiority over the other five competitors. Specifically, when the community structures are relatively clear and distinct ($\xi \leq 0.3$), most algorithms perform relatively well. In particular, FMHMO can achieve four metric values larger than 0.9, demonstrating significant advantages over the four competitors. As ξ increases from 0.3 to 0.6, all algorithms experience a decline in metric values, while FMHMO exhibits obvious superiority in stability and accuracy. When the community structures become ambiguous ($0.6 \leq \xi \leq 0.9$), all algorithms deteriorate in four

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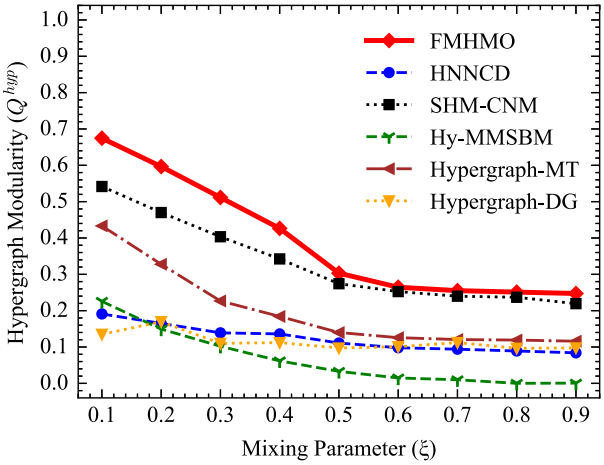


Fig. 8. Average values of hypergraph modularity Q^{hyp} derived from 20 independent runs of FMHMO and five state-of-the-art hypergraph community detection methods over a group of synthetic H-ABCD benchmarks with noise parameter ξ increased from 0.1 to 0.9.

metric values, whereas FMHMO still maintains relatively better accuracy in most test cases.

Fig. 8 presents the average values of hypergraph modularity Q^{hyp} derived from 20 independent runs of FMHMO and the five competitors on the H-ABCD hypergraph set. FMHMO consistently achieves the highest Q^{hyp} values, outperforming the five competitors across all nine H-ABCD benchmark hypergraphs. Specifically, when the community structures are relatively distinct ($\xi \leq 0.5$), all algorithms exhibit relatively better modularity quality, and FMHMO keeps obvious superiority over the other four competitors. However, as the community structures become more ambiguous (ξ increases from 0.5 to 0.9), all algorithms experience a decline in Q^{hyp} values, and FMHMO still maintains relatively better quality.

The empirical results demonstrate that the proposed FMHMO is effective in identifying accurate and high-quality communities from synthetic hypergraph benchmarks. Specifically, FMHMO demonstrates competitive performance across four accuracy metrics (i.e., NMI, ARI, F_1 and FMI) and one quality metric (i.e., Q^{hyp}) when compared to five state-of-the-art hypergraph community detection algorithms. In particular, FMHMO exhibits significant advantages in accuracy and modularity quality when community structures are relatively clear, and maintains stable when community structures become ambiguous. Furthermore, the consistent advantages observed on five different metrics prove the performance robustness of FMHMO. Among the five competitors, the SHM-CNM demonstrates superior accuracy and modularity quality. In contrast, the Hypergraph-DG and HNNCD show lower accuracy across most metrics, while the Hy-MMSBM exhibits the lowest modularity quality.

C. Performance on Real-World Hypergraphs

Second, we also assess the performance of FMHMO on real-world hypergraphs. The accuracy and modularity quality achieved by FMHMO is benchmarked against five state-of-the-art hypergraph community detection algorithms: Hypergraph-DG [13], Hypergraph-MT [16], Hy-MMSBM [4], SHM-CNM [9], and HNNCD. Ten typical real-world networks with different types and characteristics, ranging from 75 to 19 717 nodes, are analyzed, with details provided in Table I.

Table III records the average value and standard deviation of four accuracy metrics (i.e., NMI, ARI, F_1, and FMI) obtained by running FMHMO and competitors independently for 20 times on each hypergraph. We can observe that FMHMO achieves the best hypergraph partition in terms of four accuracy metrics in 20 out of 40 (50%) test instances. Specifically, in the first four smaller-scale hypergraphs (Hospital, Workspace, PrimarySchool, and HighSchool) with less than 1000 nodes, FMHMO obtains the highest metric values in 62.5% test instances. Although FMHMO underperforms HNNCD and Hypergraph-MT on the Hospital and PrimarySchool networks, it still obtains competitive performance with sub-optimal metric values. Furthermore, in the other six hypergraphs with node scale increases from 1290 to 19 717, FMHMO achieves the highest metric values in 41.7% test instances, showing superior accuracy than the five competitors on larger-scale hypergraphs.

Table IV records the average value and standard deviation of hypergraph modularity Q^{hyp} obtained by running FMHMO and the five competitors independently for 20 times on each real-world hypergraph. FMHMO obtains the highest value of Q^{hyp} on 8 out of 10 (80%) hypergraphs, demonstrating the best modularity quality among all algorithms. Although FMHMO underperforms SHM-CNM and Hy-MMSBM on two hypergraphs related to Hospital and GeneDisease, respectively. It significantly outperforms all five competitors on the other eight hypergraphs. When compared to the average Q^{hyp} value achieved by all competitors across all hypergraphs, FMHMO demonstrates an impressive average improvement of 113.0%. This significant improvement indicates the strong ability of FMHMO in enhancing partition quality identified from real-world hypergraphs.

The results in Tables III and IV are further evaluated using a typical nonparametric statistical procedure, i.e., Wilcoxon singed-rank test [48], [49], for checking the statistical difference between FMHMO and five competitors. When FMHMO is compared with each competitor, the p-value is consistently less than 0.05, demonstrating the superiority of FMHMO in achieving more accurate and higher quality partitions. In addition, the consistent advantages observed on five metrics prove the performance robustness of FMHMO. These findings prove the effectiveness of FMHMO in discovering accurate and high-quality real-world hypergraph partitions. Among the five competitors, HNNCD achieves the relatively highest accuracy by obtaining the highest metric values in 30% of the test instances, outperforming SHM-CNM and Hypergraph-MT, which achieve the highest metric values in approximately 10% of the test instances. Additionally, Hy-MMSBM and Hypergraph-DG demonstrate superior average modularity quality compared to other competitors. Moreover, SHM-CNM and Hy-MMSBM achieve the highest modularity value on the Hospital and Gene-Disease hypergraphs, respectively.

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TABLE III
AVERAGE OPTIMAL VALUE AND STANDARD DEVIATION OF FOUR METRICS (NMI, ARI, F_1, AND FMI) OBTAINED BY FMHMO AND FIVE STATE-OF-THE-ART HYPERGRAPH COMMUNITY DETECTION ALGORITHMS PERFORMING 20 INDEPENDENT RUNS ON TEN REAL-WORLD HYPERGRAPHS

Hypergraph	Metrics	Hypergraph-DG [13] Mean(Std.)	Hypergraph-MT [16] Mean(Std.)	Hy-MMSBM [4] Mean(Std.)	SHM-CNM [9] Mean(Std.)	HNNCD Mean(Std.)	FMHMO Mean(Std.)
Hospital	NMI	0.143(0.00E+00)	0.030(6.53E-03)	0.132(1.60E-02)	0.045(0.00E+00)	0.098(4.59E-02)	**0.145(0.00E+00)**
	ARI	0.056(0.00E+00)	0.019(1.22E-02)	0.064(2.07E-02)	0.003(0.00E+00)	**0.074(4.34E-02)**	0.071(0.00E+00)
	F1	0.253(0.00E+00)	0.415(9.33E-03)	0.483(9.98E-03)	0.413(0.00E+00)	**0.507(5.04E-02)**	0.453(0.00E+00)
	FMI	0.198(0.00E+00)	0.457(2.93E-02)	0.488(2.22E-02)	0.407(2.71E-02)	0.443(5.55E-02)	**0.524(0.00E+00)**
Workspace	NMI	0.686(0.00E+00)	0.760(1.68E-02)	0.584(5.82E-02)	0.754(0.00E+00)	0.224(6.93E-02)	**0.836(0.00E+00)**
	ARI	0.681(0.00E+00)	0.680(1.72E-02)	0.488(1.21E-01)	0.759(0.00E+00)	0.180(6.82E-02)	**0.849(0.00E+00)**
	F1	0.859(0.00E+00)	0.793(2.53E-02)	0.620(8.98E-02)	0.880(0.00E+00)	0.528(5.96E-02)	**0.924(0.00E+00)**
	FMI	0.772(0.00E+00)	0.758(1.34E-02)	0.615(9.14E-02)	0.694(5.58E-02)	0.472(5.07E-02)	**0.891(0.00E+00)**
PrimarySchool	NMI	0.682(0.00E+00)	**0.748(1.80E-02)**	0.633(3.02E-02)	0.378(0.00E+00)	0.213(7.00E-02)	0.718(0.00E+00)
	ARI	0.504(0.00E+00)	**0.637(3.90E-02)**	0.427(5.51E-02)	0.200(0.00E+00)	0.096(4.65E-02)	0.545(0.00E+00)
	F1	0.649(0.00E+00)	**0.783(3.31E-02)**	0.542(5.62E-02)	0.306(0.00E+00)	0.269(6.97E-02)	0.678(0.00E+00)
	FMI	0.568(0.00E+00)	**0.672(3.37E-02)**	0.490(4.89E-02)	0.389(0.00E+00)	0.294(1.27E-02)	0.606(0.00E+00)
HighSchool	NMI	0.853(0.00E+00)	0.458(1.09E-02)	0.707(4.09E-02)	0.627(0.00E+00)	0.254(8.09E-02)	**0.959(3.30E-03)**
	ARI	0.820(0.00E+00)	0.192(1.55E-02)	0.519(4.25E-02)	0.369(0.00E+00)	0.152(7.40E-02)	**0.957(3.90E-03)**
	F1	0.896(0.00E+00)	0.257(0.00E+00)	0.544(3.33E-02)	0.370(0.00E+00)	0.337(9.10E-02)	**0.980(1.96E-03)**
	FMI	0.839(0.00E+00)	0.452(9.13E-03)	0.579(3.92E-02)	0.553(0.00E+00)	0.344(2.64E-02)	**0.961(3.46E-03)**
HouseCommittees	NMI	0.020(0.00E+00)	0.005(8.19E-04)	0.006(1.34E-03)	0.104(1.25E-03)	**0.863(1.73E-02)**	0.010(3.36E-04)
	ARI	0.001(0.00E+00)	0.000(2.48E-04)	0.000(5.21E-04)	0.002(2.20E-05)	0.000(5.34E-04)	**0.005(5.78E-04)**
	F1	0.061(0.00E+00)	0.056(4.00E-03)	0.101(8.66E-03)	0.318(2.71E-03)	**0.513(4.95E-03)**	0.343(1.00E-01)
	FMI	0.118(0.00E+00)	0.142(6.26E-04)	0.172(5.23E-03)	0.385(3.84E-03)	**0.503(9.54E-04)**	0.404(1.67E-01)
Cora_ca	NMI	0.409(0.00E+00)	0.093(1.29E-02)	0.102(1.70E-02)	0.405(2.06E-02)	0.015(6.83E-03)	**0.452(7.12E-03)**
	ARI	0.008(0.00E+00)	0.054(7.84E-03)	0.060(1.36E-02)	0.244(2.38E-02)	-0.003(3.28E-03)	**0.249(7.66E-03)**
	F1	0.024(0.00E+00)	0.292(1.47E-02)	0.293(2.04E-02)	0.432(2.58E-02)	0.244(1.15E-02)	**0.446(8.91E-03)**
	FMI	0.065(0.00E+00)	0.203(6.55E-03)	0.212(1.28E-02)	**0.381(1.91E-02)**	0.314(4.20E-03)	0.347(7.01E-03)
Citeseer	NMI	0.344(0.00E+00)	0.016(1.70E-02)	0.038(9.47E-03)	0.343(2.52E-03)	0.021(6.92E-03)	**0.350(1.57E-03)**
	ARI	0.008(0.00E+00)	0.006(1.02E-02)	0.028(7.19E-03)	0.140(7.32E-03)	0.015(5.53E-03)	**0.142(3.27E-03)**
	F1	0.028(0.00E+00)	0.187(2.60E-02)	0.260(1.78E-02)	0.225(1.23E-02)	0.261(1.19E-02)	**0.269(5.37E-03)**
	FMI	0.060(0.00E+00)	0.167(1.06E-02)	0.201(7.24E-03)	0.263(8.00E-03)	**0.306(3.48E-02)**	0.255(3.16E-03)
EnronEmail	NMI	0.021(0.00E+00)	0.002(2.57E-03)	0.001(1.36E-03)	0.012(1.10E-03)	**0.214(4.35E-02)**	0.005(3.00E-06)
	ARI	0.000(0.00E+00)	0.003(2.77E-03)	0.002(3.87E-03)	0.004(1.54E-03)	**0.319(4.07E-02)**	0.004(5.90E-05)
	F1	0.007(0.00E+00)	0.581(5.24E-02)	0.628(7.01E-02)	0.348(3.96E-02)	**0.961(2.14E-02)**	0.338(2.82E-04)
	FMI	0.114(2.53E-01)	0.700(1.77E-02)	0.718(2.41E-02)	0.430(2.56E-02)	**0.961(2.20E-02)**	0.412(7.92E-04)
GeneDisease	NMI	0.171(0.00E+00)	/	0.006(1.71E-03)	**0.399(1.86E-03)**	0.069(9.45E-04)	0.028(4.59E-04)
	ARI	0.003(0.00E+00)	/	0.000(1.70E-04)	**0.020(1.50E-03)**	0.007(2.53E-04)	0.001(1.30E-05)
	F1	0.041(0.00E+00)	/	0.070(1.26E-03)	0.051(1.33E-03)	**0.077(3.76E-04)**	0.075(1.59E-04)
	FMI	0.029(0.00E+00)	/	0.162(3.12E-03)	0.134(2.85E-03)	0.188(1.23E-03)	**0.200(1.53E-04)**
Pubmed	NMI	0.209(0.00E+00)	0.001(3.55E-04)	0.086(2.18E-02)	0.162(9.67E-03)	0.006(7.50E-03)	**0.253(3.22E-03)**
	ARI	0.005(0.00E+00)	0.000(4.52E-04)	0.100(2.47E-02)	0.187(3.54E-02)	0.006(8.17E-03)	**0.222(1.39E-02)**
	F1	0.034(0.00E+00)	0.345(6.08E-03)	**0.498(2.17E-02)**	0.416(2.69E-02)	0.419(9.01E-03)	0.404(2.32E-02)
	FMI	0.063(0.00E+00)	0.346(3.20E-04)	0.414(1.68E-02)	0.405(3.17E-02)	**0.419(1.27E-02)**	0.418(1.41E-02)
p-value		4.00E-06	9.20E-05	4.40E-05	1.55E-02	1.93E-02	-

Note that "/" means that no effective hypergraph partition can be obtained. The best solutions obtained by all algorithms on each hypergraph is shown in bold.

TABLE IV
AVERAGE OPTIMAL VALUE AND STANDARD DEVIATION OF HYPERGRAPH MODULARITY Q^{hyp} OBTAINED BY FMHMO AND FIVE STATE-OF-THE-ART HYPERGRAPH COMMUNITY DETECTION ALGORITHMS PERFORMING 20 INDEPENDENT RUNS ON TEN REAL-WORLD HYPERGRAPH

Hypergraph	Hypergraph-DG [13] Mean(Std.)	Hypergraph-MT [16] Mean(Std.)	Hy-MMSBM [4] Mean(Std.)	SHM-CNM [9] Mean(Std.)	HNNCD Mean(Std.)	FMHMO Mean(Std.)
Hospital	0.054(0.00E+00)	0.013(3.41E-03)	0.088(3.28E-03)	**0.093(0.00E+00)**	0.007(1.09E-02)	0.080(0.00E+00)
WorkSpace	0.340(0.00E+00)	0.356(1.30E-02)	0.224(4.94E-02)	0.371(0.00E+00)	0.135(3.17E-02)	**0.394(0.00E+00)**
PrimarySchool	0.406(0.00E+00)	0.383(1.61E-03)	0.294(2.89E-02)	0.328(0.00E+00)	0.127(3.38E-02)	**0.422(0.00E+00)**
HighSchool	0.577(0.00E+00)	0.411(4.35E-03)	0.299(2.72E-02)	0.558(0.00E+00)	0.226(5.20E-02)	**0.623(6.19E-04)**
HouseCommittees	0.188(0.00E+00)	-0.001(4.77E-03)	0.169(1.10E+02)	0.024(2.55E-03)	0.037(8.90E-03)	**0.314(2.18E-03)**
Cora_ca	0.331(0.00E+00)	0.525(3.53E-03)	0.269(4.29E-02)	0.672(4.78E-03)	0.027(3.44E-02)	**0.755(4.35E-03)**
Citeseer	0.500(0.00E+00)	0.174(2.26E-02)	0.247(7.61E-02)	0.854(4.55E-03)	0.166(5.87E-02)	**0.883(3.34E-04)**
EnronEmail	0.142(0.00E+00)	0.351(3.70E-02)	0.375(1.70E-02)	0.565(1.50E-02)	-0.140(2.96E-02)	**0.603(2.03E-04)**
GeneDisease	0.205(0.00E+00)	/	**0.305(7.35E-03)**	0.056(3.70E-03)	0.041(1.06E-02)	0.135(5.11E-04)
Pubemd	0.266(0.00E+00)	2.63E-04(4.48E-03)	0.383(1.31E-01)	/	0.017(2.50E-02)	**0.653(3.39E-04)**
p-value	2.18E-02	5.06E-03	2.48E-02	6.91E-03	5.06E-03	-

Note that "/" means that no effective hypergraph partition can be obtained. A negative value indicates a node pair from the same community participates in lesser than the expected number of hyperedges. The best solutions obtained by all algorithms on each hypergraph is shown in bold.

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TABLE V
AVERAGE RUNTIME REQUIRED BY FMHMO AND FIVE STATE-OF-THE-ART HYPERGRAPH COMMUNITY DETECTION ALGORITHMS ON FIVE SYNTHETIC H-ABCD BENCHMARKS

Hypergraphs	Hypergraph-DG [13] Runtime(s)	Hypergraph-MT [16] Runtime(s)	Hy-MMSBM [4] Runtime(s)	SHM-CNM [9] Runtime(s)	HNNCD Runtime(s)	FMHMO Runtime(s)
H-ABCD_1000	**1.36**	16.65	3.08	366900.00	23.01	26.77
H-ABCD_2000	**3.27**	48.97	4.98	/	73.00	107.41
H-ABCD_3000	**6.53**	77.62	6.78	/	104.16	138.70
H-ABCD_4000	15.56	153.38	**12.36**	/	209.19	544.42
H-ABCD_5000	21.48	244.42	**17.80**	/	308.46	976.37

Note that "/" means that no effective results can be obtained within 48 hours. The best solutions are shown in bold.

The results on both synthetic and real-world hypergraphs indicate that the proposed FMHMO is competitive in identifying accurate and high-quality hypergraph partitions, outperforming four state-of-the-art hypergraph community detection algorithms. Specially, the performance superiority maintains stable on five types of performance metrics. The success of FMHMO primarily stems from the utilization of fine-grained hypergraph fuzzy memberships, which effectively enhances performance in maintaining higher-order hyperedge information and accurately recovering hypergraph partitions. The hypergraph fuzzy memberships can also be applied to other approaches for hypergraph community detection.

D. Time Efficiency Analysis of FMHMO

At last, the time efficiency of FMHMO is compared with the five hypergraph community detection algorithms: Hypergraph-DG [13], Hypergraph-MT [16], Hy-MMSBM [4], SHM-CNM [9], and HNNCD. A group of five synthetic H-ABCD benchmarks with the scale increased from 1000 to 5000 nodes are adopted, and the noise parameter ξ is set at 0.5.

Table V records the average value of runtime (s) required by FMHMO and five competitors in 20 independent runs on each hypergraph. It can be observed that the time efficiency of FMHMO is generally acceptable, demonstrating a certain level of scalability. Specifically, when the scale of the synthetic H-ABCD benchmarks increases to 5000 nodes, the total runtime for FMHMO remains under 1000 s. Although FMHMO may not match the efficiency of four competitors, i.e., the derivative graph-based algorithm (Hypergraph-DG) and probabilistic model-based algorithms (Hy-MMSBM and Hypergraph-MT), and HNN-based algorithm (HNNCD), it still demonstrates superior time efficiency relative to the SHM-CNM algorithm, which relies on hypergraph modularity optimization. To further enhance time efficiency when encountering large-scale hypergraphs, heuristic or metaheuristic frameworks based on parallel computing [50] and machine learning [51], [52] can be employed.

VI. CONCLUSION

In summary, we define novel concepts of multi-scale hypergraph fuzzy memberships to systematically quantify the partial belongingness (i.e., membership grades) among nodes, hyperedges, and hypergraph communities within the interval $[0, 1]$. These fuzzy memberships provide richer and finer-grained fuzzy community information for real-life hypergraphs. Moreover, we utilize the multiscale hypergraph fuzzy memberships to enhance crisp hypergraph community detection via a general framework of FMHMO. Compared with state-of-the-art hypergraph community detection methods, FMHMO significantly improves the accuracy and modularity quality of the identified crisp hypergraph partitions of nodes in both synthetic and real-world datasets.

First, the multiscale hypergraph fuzzy memberships are established based on the higher-order hyperedges, encompassing node-to-hyperedge, hyperedge-to-community, and node-to-community memberships. Second, the FMHMO framework incorporates the multiscale hypergraph fuzzy memberships as fine-grained neighbor fuzzy information to enhance hypergraph modularity optimization, improving the identified crisp hypergraph partitions. Specifically, FMHMO consists of two primary strategies: hyperedge-similarity-based hypergraph reduction (HS-HR) and FM-HPR. Notably, HS-HR simplifies a hypergraph into a weighted graph, with edge weights determined by a novel measure of hyperedge similarity, thereby preserving of both original intrahyperedge and interhyperedge topologies. Meanwhile, FM-HPR recovers a hypergraph partition from an approximately modularity-optimized weighted graph partition. It evaluates the fuzzy memberships of hyperedge-based bridge nodes, thus facilitating the accurate determination of their crisp hypergraph community affiliations.

This work highlights the significance of multi-scale hypergraph fuzzy memberships in enriching fuzzy community information for hypergraphs, as well as their potential in assisting modularity optimization for enhancing crisp hypergraph partitions. These fuzzy memberships provide a fresh perspective on understanding meso-scale community structures in real-world hypergraphs and the topological characteristics of critical higher-order structural units, i.e., hyperedges. High-quality hypergraph partitions with multi-scale fuzzy memberships also offer valuable insights for practical hypergraph applications, such as influence maximization, link prediction, and collaborative filtering. In future research, we will strive to incorporate an iterative approach in FMHMO to further enhance the hypergraph community detection performance while reducing time complexity.

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